Mustafa Mehboob

Product Engineer · Security · Payments · Cloud

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Summary

Full-stack software engineer with **3+ years** on products that take payments and hold personal data. Most recently took over a UK visa platform during a live cyber attack, stabilised it, rebuilt its AWS architecture, then **led its three-engineer team as revenue passed $1M** and launched a second brand on the same platform. **First author of a 2026 paper on LLM-based software testing and vulnerability detection**.

Experience

Software Engineer Jan 2026 – Present
Softaims *Lahore, Pakistan*

- **Took over UKVisaPortal during a live attack**, after a data breach. Contained the incident, traced the entry route and closed the whole class of weakness rather than the single hole. The site stayed up through a Layer-7 DDoS that peaked at 700M requests, **∼600M of them blocked**.
- Rebuilt the AWS architecture around one rule: traffic reaches the application only through CloudFront and the WAF. Locked the origin, moved workloads to ECS Fargate behind a private load balancer, and encrypted passport scans with AES-256-GCM before they are written to S3. Moved the platform off end-of-life Node 20; the image build now refuses an unpatched runtime.
- **Led the three-engineer team** after the recovery: built the admin suite staff use to process applications (call queues, auto-assignment, productivity reports, GDPR erasure and retention), integrated Adyen, MamoPay and PayPal, and built staff sign-in with authenticator-app two-factor and passkeys (WebAuthn). **Revenue passed $1M.**
- Launched a second brand, Crown Visa, on the same platform: each brand ships as its own build, which fails if the other brand's text leaks in, and payments, emails and refunds follow the brand and domain a customer used.
- **Co-led WorldVisaHub**, a multi-country visa platform where supervisors build each country's application form themselves. Made published form versions immutable and pinned to in-flight orders, so an edit never changes an application in progress.
- Designed its document pipeline with **no long-lived cloud keys**: uploads go straight to S3 on short-lived STS credentials, files are checked by content rather than extension, encrypted with KMS, and personal data is scrubbed from logs. Provisioned staging in Terraform, with keyless OIDC deploys gated on Brakeman and bundler-audit.
- Building an industrial grain-aeration SCADA backend (NestJS, OPC UA) in which one service holds every site's PLC session: connections open in stages with jittered retries, every node is re-read after an outage, and each organisation's admin manages only its own PLCs. Deployed to AWS ECS through Terraform and CI; verified against the client's test PLC over the site VPN.

Associate Software Engineer Jun 2025 – Dec 2025
OptimaGeeks *Lahore, Pakistan*

- **Led Userbird**, a GA- and PostHog-style analytics and session-replay platform: an embeddable JavaScript SDK streaming pageviews, events and recordings into BigQuery. Added caching and lazy loading that made the dashboard **40% faster**.
- Shipped production apps including a LinkedIn Sales Navigator to CRM extension, OpenAI-powered CRM tooling, WordPress publishing automation, a bot on Cloudflare Workers and a NestJS backend.

Associate Software Developer Mar 2023 – Feb 2025
Webmended (UAE) *Remote*

- **Led mDrive Asia** end to end: storefront, checkout, admin and fulfilment, including Apple GSX and custom payment integrations (Next.js, Django, PostgreSQL). Also built a Django quality-management system and PHP client projects.

LAMP Stack Developer Feb 2022 – Mar 2023
easybew (Canada) *Remote*

Publications

First author. "AI-Assisted Automated Software Testing and Vulnerability Detection Using Large Language Models." *Spectrum of Engineering Sciences* 4(3), 2026.
Co-author. "Transfer Learning Optimization for Low-Resource Natural Language Processing Using Transformer Architectures." *Research Consortium Archive* 4(2), 2026.

Skills

Build: TypeScript, Node.js (NestJS, Express), React, Next.js, Python (Django), Ruby on Rails, OpenAI API
Data: PostgreSQL, Redis, MongoDB, BigQuery, SQL
Run: AWS (ECS Fargate, RDS, S3, CloudFront, WAF, KMS, IAM/STS), Terraform, Docker, GitHub Actions, Cloudflare
Secure: OAuth 2.0, Microsoft SSO, RBAC, TOTP and WebAuthn passkeys, encryption at rest, WAF and DDoS response, GDPR

Education

Government College University (GCU) Lahore Nov 2021 – May 2025
Bachelor of Science in Computer Science (BSCS), CGPA 3.18 *Lahore, Pakistan*